

EDUCATION

GEORGETOWN UNIVERSITY

M.S. in Data Science & Analytics

Washington, DC

Expected May 2027 | GPA: 4.0/4.0

- Teaching Assistant: Database Systems and SQL; AI Modeling

UNIVERSIDAD IBEROAMERICANA

Mexico City, MX

Bachelor's degree in Mechatronics and Production Engineering

- Honors: Awarded Tuition Scholarship for academic merit

WORK EXPERIENCE

WELO DATA

Mexico City, MX

Data Science & Analytics Engineer

Nov 2024 – Aug 2025

- Developed XGBoost model predicting task difficulty across 120k+ records, improving annotation accuracy by 10% via optimized task allocation
- Developed GMM-based recommendation system for rater-project matching, improving annotation quality by 15% and 10% across two projects
- Automated financial ETL pipeline in Redshift processing high-volume transactions (0.001% error rate), reducing reconciliation time by 70%
- Led analytics for two projects, building end-to-end ML pipelines for real-time KPI monitoring

SCALE AI

Mexico City, MX

Data Operations Engineer

Oct 2022 – Nov 2024

- Developed automated evaluation frameworks for NVIDIA Nemotron's RLHF alignment using GPT-4 as Judge-LLM and HelpSteer2 methodology
- Built LLM quality linting system adopted across GenAI organization, automating error detection via programmatic flagging and reducing manual review by 40%
- Built TypeScript-MongoDB integrations and Python pipelines to automate end-to-end customer data processing

Operations Data Analyst & Specialist

Feb 2021 – Oct 2022

- Built Logistic Regression model on 12+ months of operational data to predict batch acceptance, improving approval rates by 20%

PROJECTS

ModelLens — NN Interpretability Toolkit ([GitHub](#))

Georgetown University - In Progress

- Built an architecture-agnostic PyTorch interpretability toolkit (transformers, CNNs, LSTMs, MLPs) with a unified API (activation patching, circuit discovery, logit lens, residual-stream, attention maps) via an auto-detecting adapter layer
- Applied it to a behavior-controlled circuit comparison showing activation steering and LoRA produce the same output through different circuits - a concentrated mid-layer MLP pathway vs. a distributed multi-layer effect

Stoic LLM Steering ([GitHub](#))

Personal Project - In Progress

- Built an end-to-end steering pipeline (Llama-3.2, 1B→3B) using Contrastive Activation Addition (CAA) and LoRA fine-tuning, Claude-API contrastive-pair generation, layer/coefficient sweeps, and LLM-as-judge evaluation
- Characterized the limits of activation steering across three instruments (style, content, decision) with seed-averaged, decoding-matched probes, showing the Stoic vector reliably shifts register, not decisions

Political Sentiment Analysis ([GitHub](#))

Georgetown University

- Developed DistilBERT-based NLP pipeline for emotion classification across 2,000+ Congressional speeches, identifying statistically significant partisan patterns

PUBLICATIONS

Villalobos-Alva, S., et al. (2022). Protein Science Meets AI: Systematic Review. Front. Bioeng. Biotechnol. [\[Link\]](#)

SKILLS & OTHER INFORMATION

AI & Machine Learning: Generative AI (LLMs, RLHF, HelpSteer2), Mechanistic Interpretability (Activation Patching, Logit Lens, Residual Stream Analysis), NLP/NLU, Computer Vision, Deep Learning, LLM Evaluation (Judge-LLM), XGBoost, GMM
Programming: Python (Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Hugging Face), SQL, NoSQL, TypeScript, R, Git, GitHub, Linux/CLI

Languages & Certifications: Spanish (Native), English (Fluent); MIT MicroMasters in Statistics and Data Science